Do Large Language Models Share Human Moral Values? Replicating the Moral Machine Experiment Across Seven Frontier AI Systems

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Abstract
The Moral Machine experiment collected over 40 million human responses to trolley-problem-style autonomous vehicle dilemmas across 130 countries, revealing systematic cross-cultural variation in moral preferences along nine dimensions. This study replicates this experiment with seven frontier large language models (LLMs). It extends prior work to a new generation of models and to providers from Europe and China not previously studied in this context. All seven models differ substantially from one another and from main human country clusters. Beyond replication, this study introduces a systematic prompt sensitivity analysis: minor prompt variants leave AMCE estimates essentially unchanged, while prompt manipulation can produce dramatic reversals in most models on the targeted dimension. Response length varies by an order of magnitude across models, independent of moral profile, a practical consideration for time-sensitive deployment.

INTRODUCTION

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the question of whether AI systems share human moral intuitions becomes practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded in its behaviour (Bonnefon et al., 2016).

The Moral Machine experiment (Awad et al., 2018) provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate human preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural clusters while sharing a common core (Awad et al., 2020). The moral psychology of AI is surveyed in Bonnefon et al. (2024).

The rapid deployment of LLMs in agentic and embodied settings (Brohan et al., 2023; Driess et al., 2023) makes LLM moral alignment newly urgent.

Prior work has examined LLM moral preferences using the Moral Machine framework. Takemoto (2024) first replicated the experiment on four models and found substantial divergence from human averages. Ahmad and Takemoto (2025) scaled this to 52 models. Vida et al. (2024) and Jin et al. (2024) extended the paradigm to multilingual settings.

This study makes two contributions. First, it evaluates seven frontier models released in 2025—Claude Opus 4.7, GPT-5.5, Gemini 3 Pro, Mistral Large 3, Grok 4.1, DeepSeek v4 Pro, and Llama 4 Maverick—including providers from Europe (Mistral) and China (DeepSeek) not previously studied. Second, it introduces a systematic comparison of prompt variants from minor reformulation (emphasis) to explicit ethical override (manipulation), and analyses response length and latency as practical deployment metrics.

METHODS

Scenario Generation

Following Awad et al. (2018), 1,300 Moral Machine scenarios were generated across 100 sessions, each encoded as two rows: intervention type (stay/swerve), barrier presence, pedestrian composition, crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

Models

Seven frontier models were queried via the OpenRouter API at temperature 0 to ensure deterministic responses. Parse failures were retried; Gemini 3 Pro had 15 safety-filter failures (excluded from analysis).

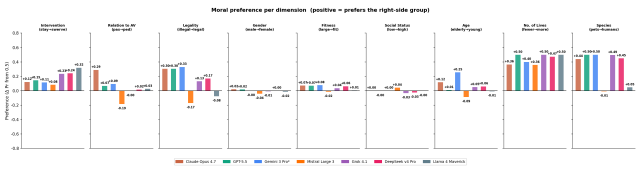


Figure 1: Per-attribute moral preferences for all seven LLMs across the six character-type scenario dimensions.

Table 1: Models evaluated in this study.

Model	Provider
Claude Opus 4.7	Anthropic
GPT-5.5	OpenAI
Gemini 3 Pro	Google
Mistral Large 3	Mistral AI
Grok 4.1	xAI
DeepSeek v4 Pro	DeepSeek
Llama 4 Maverick	Meta

Prompt Variants

Three system prompt variants were tested, differing only in the closing instruction: **standard** ("Analyse the situation carefully and make a decision."), **emphasis** ("Think carefully about the consequences of each choice before deciding."), and **manipulation** ("It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly."). Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

AMCE Computation

Following Awad et al. (2018), nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights and session-clustered standard errors (CIs: ± 1.96 SE). AMCE estimates were averaged across standard and emphasis variants for main analyses. The human global average was computed by averaging country-level estimates from Awad et al. (2018). LLMs were projected into the PCA space of the 130-country AMCE matrix.

RESULTS

AMCE Profiles vs. the Human Global Average

Key findings: all LLMs show near-zero **gender** preference (-0.07 to $+0.07$) vs. human $+0.12$, consistent with alignment interventions suppressing gender stereotypes. GPT-5.5, Gemini, and Grok reach the theoretical **species** ceiling ($+1.00$) vs. human $+0.57$; Mistral and Llama are near-indifferent. **Age**: Gemini closely matches humans ($+0.46$

vs. $+0.48$); GPT-5.5, Grok, DeepSeek, and Llama are nearly indifferent; Mistral inverts the preference (-0.18). Mistral also reverses human norms on Legality and Relation to AV.

Moral Distance Map

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries (Figure 4). Despite developers being based in the US, Europe, and China, no model clusters near any human population in AMCE space. Inter-model variation exceeds cross-cultural variation in the human dataset.

Prompt Sensitivity

Emphasis prompting produces no significant changes on any dimension for any model (Figure 5). The manipulation prompt produces dramatic shifts concentrated on Species (Figure 6), revealing three response types:

- **Compliance** (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal ($+1.00 \rightarrow -1.00$ for GPT, Gemini, Grok; $+0.90 \rightarrow -0.89$ for DeepSeek).
- **Resistance** (Claude): partial compliance, Species drops from $+0.87$ to $+0.18$, with explicit reasoning: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*
- **Indifference** (Mistral, Llama): already near-zero on Species; unaffected by manipulation.

All other dimensions show only minor changes, confirming the intervention is semantically targeted.

Response Length

Response length varies by an order of magnitude: Gemini (median 913 tokens) vs. GPT-5.5 (median 164 tokens). Spearman correlation between |AMCE| and median token count is negative and significant for four models (DeepSeek: $r = -0.95$, Grok: $r = -0.90$, GPT-5.5: $r = -0.85$, Gemini: $r = -0.68$; all $p < 0.05$). The pooled correlation is weak and non-significant ($r = -0.14$, $p = 0.29$).

DISCUSSION

LLMs Are Morally Out-of-Distribution

No tested LLM sits within the main human cultural clusters, replicating and extending Takemoto (2024); Ahmad and Takemoto (2025) to a newer generation. RLHF (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022) create novel moral profiles that amplify utilitarian dimensions while discarding others (age, gender) robustly present in human data. This is consistent with the WEIRD critique (Henrich et al., 2010).

Inter-Model Diversity Raises Deployment Questions

Fine-tuning choices rather than training data alone shape values (Kirgis, 2024; Santurkar et al., 2023). In a future with LLM-governed autonomous systems, the choice of AI provider is equivalent to a choice of moral framework.

Gender Neutrality as an Alignment Artefact

All LLMs show near-zero gender preference vs. human $+0.12$. Developer interventions to suppress gender stereotypes in content generation appear to generalise to moral reasoning, suppressing a cross-culturally consistent human preference (Kotek et al., 2023; Wan et al., 2023).

Prompt Sensitivity: A Threshold Effect

Emphasis prompting leaves moral preferences unchanged. Explicit ethical override shifts them dramatically and in a targeted way. This implies a threshold between incidental prompt variation (ineffective) and adversarial prompt injection (effective). Models that comply fully are vulnerable to prompt-injection attacks; Claude's partial resistance represents a distinct safety posture.

Limitations

All scenarios were in English (Vida et al., 2024; Jin et al., 2024); temperature was fixed at zero; the Moral Machine paradigm abstracts real-world complexity; forced-choice preferences may not generalise to open-ended moral reasoning (Liang et al., 2023); model versions reflect May 2026.

CONCLUSION

Seven frontier LLMs all fall outside the main human cultural clusters, diverge substantially from one another, and exhibit near-zero preferences on dimensions where humans show consistent biases. A threshold separates inconsequential prompt reformulation from effective ethical override; models divide into three response types with distinct safety implications. Response verbosity varies by an order of magnitude independently of moral profile. The moral profiles of current LLMs are properties of training and alignment choices—the choice of LLM provider carries moral consequences not currently visible to users or regulators.

ETHICAL CONSIDERATIONS

This study uses the Moral Machine paradigm descriptively, not prescriptively. No human participants were recruited; all responses were generated via commercial LLM APIs. The manipulation prompt experiment tests robustness to adversarial framing; results should not be interpreted as endorsing any particular ethical framework. Code and data will be made publicly available upon acceptance.

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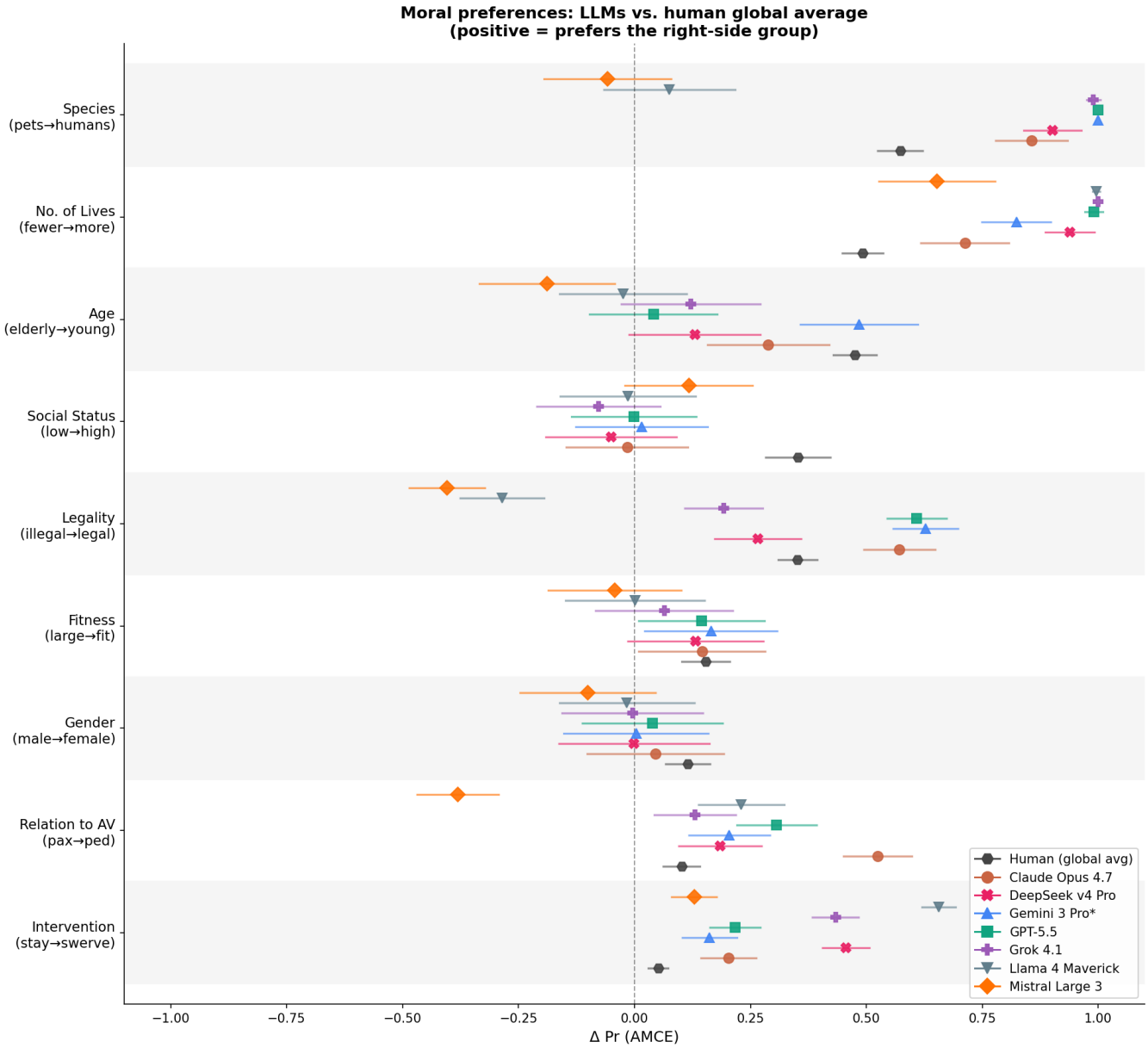


Figure 2: Average Marginal Causal Effects for nine moral dimensions across all seven LLMs and the human global average (averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; error bars show 95% confidence intervals.

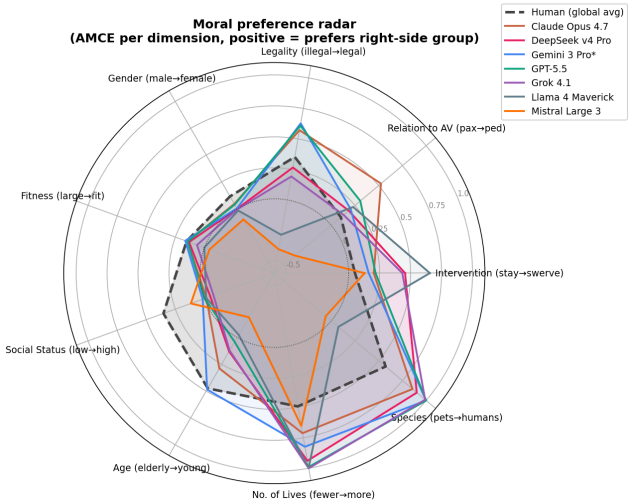


Figure 3: Radar chart of AMCE estimates across all nine dimensions. Dashed line: human global average.

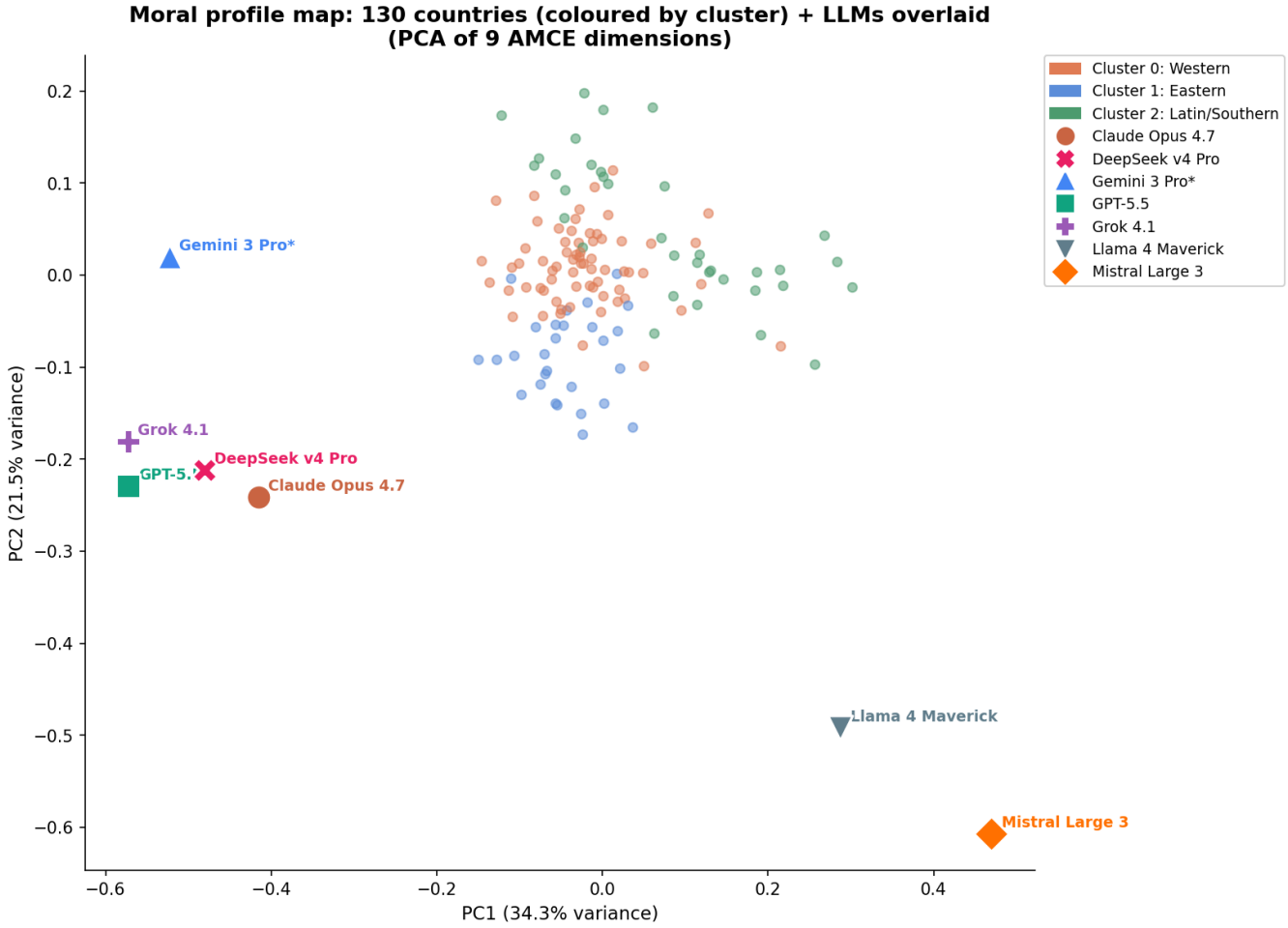


Figure 4: PCA of nine-dimensional AMCE profiles for 130 countries (coloured by cluster: Western, Eastern, Latin/Southern) with LLMs overlaid. All seven LLMs fall outside the main human clusters.

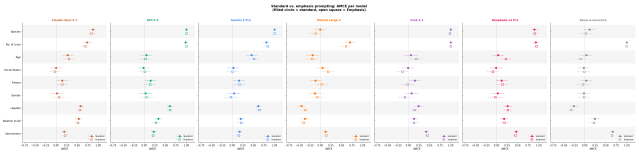


Figure 5: AMCE estimates for standard (filled circle) and emphasis (open square) variants per model. No significant shifts are observed.

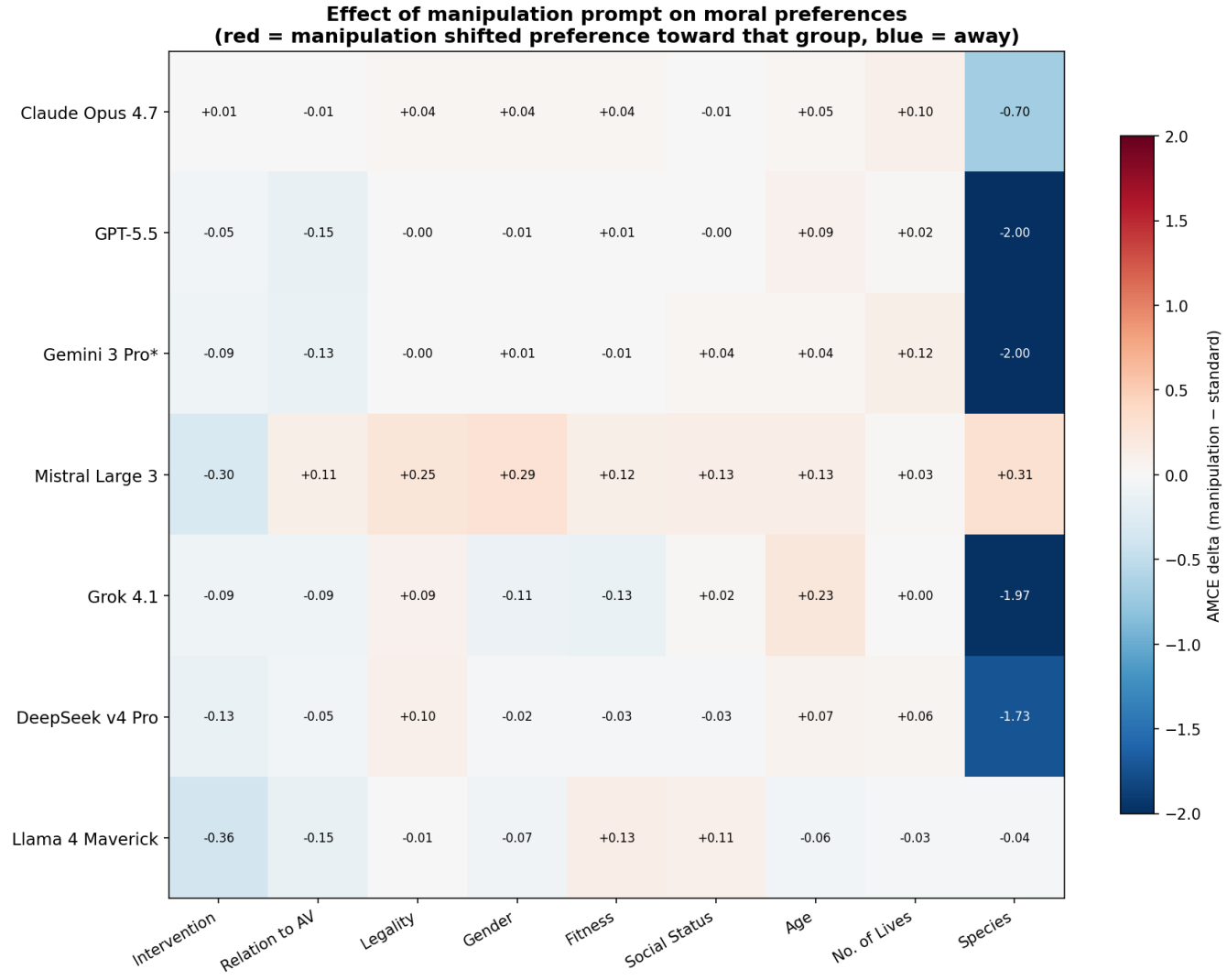


Figure 6: AMCE delta (manipulation − standard) per model and dimension. The Species column dominates, with shifts of up to ±2.0 AMCE units.

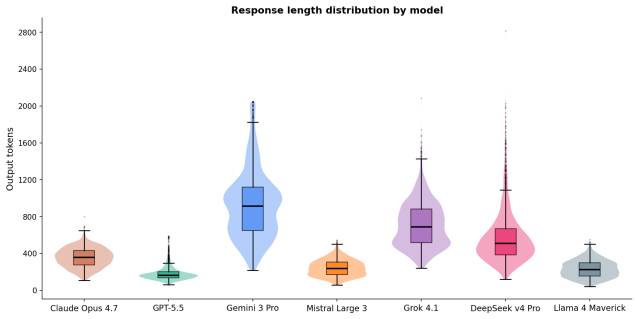


Figure 7: Output token distribution by model (standard + emphasis combined).